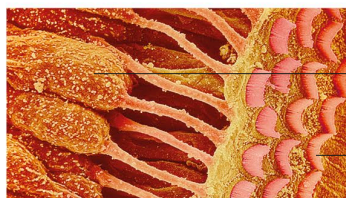


HOW WE HEAR

Ears act as energy converters, changing pressure differences in air, known as sound waves, into electrochemical nerve impulses. Sound waves usually occur as a complex pattern of frequencies, and they vibrate the eardrum in the same pattern. The vibrations are conducted along the ossicle chain, which rocks like a bent lever and forces the footplate of the stapes to act like a piston, pushing and pulling at the flexible oval window of the cochlea. The motions set off waves through the perilymph fluid inside the cochlea. These, in turn, transfer their vibrational energy to the organ of Corti, which coils within the cochlea.



Hair cell

Hairs of hair cell

HAIR CELLS

Within the organ of Corti, with the tectorial membrane removed on the right, each hair cell is seen to have 40–100 hairs arranged in a curve. Nerve fibres run from the cell bases.

Organ of Corti

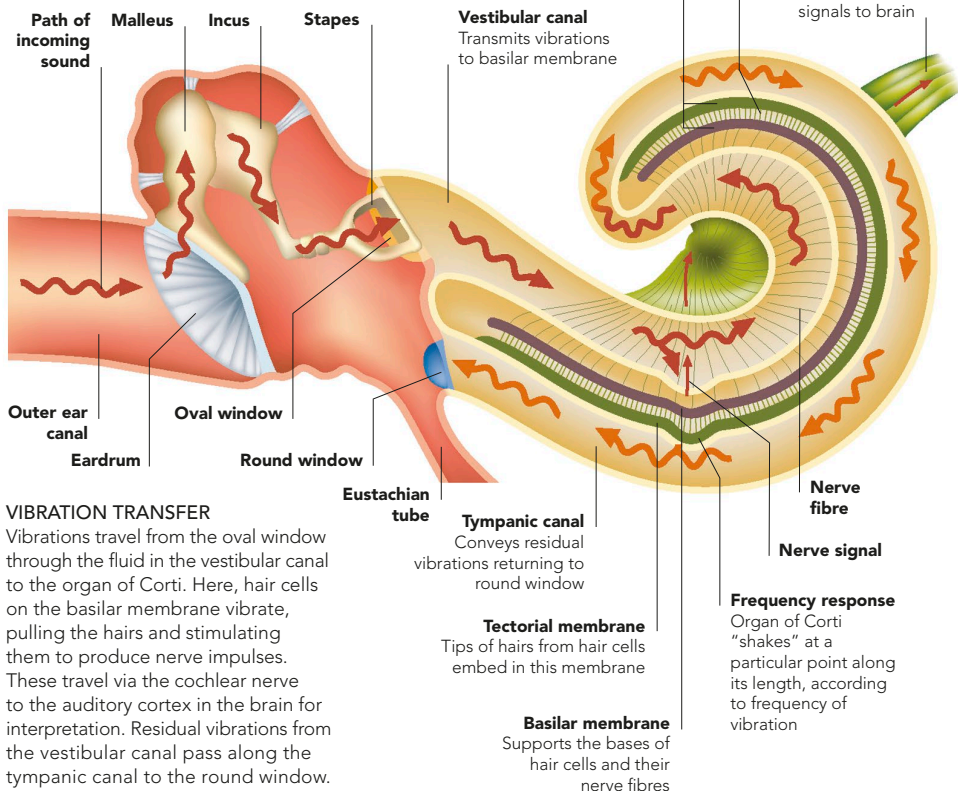
Central spiral element of the cochlea, composed of tectorial and basilar membranes linked by sensitive hair cells

Hair cells

Generate nerve signals in response to motion of basilar and tectorial membranes

Cochlear nerve

Carries nerve signals to brain



VIBRATION TRANSFER

Vibrations travel from the oval window through the fluid in the vestibular canal to the organ of Corti. Here, hair cells on the basilar membrane vibrate, pulling the hairs and stimulating them to produce nerve impulses. These travel via the cochlear nerve to the auditory cortex in the brain for interpretation. Residual vibrations from the vestibular canal pass along the tympanic canal to the round window.